

OS disk types (managed and ephemeral)

- **Managed**
 - Virtual hard disk in Azure, typically used for persistent storage
 - Azure automatically copies the operating system disk for the VM to Azure storage
 - May result in slower node provisioning and higher read/write latency
- **Ephemeral**
 - Temporary disk that is created and managed by Azure on the local storage of the virtual machine host
 - It serves as a repository for a virtual machine's boot and operating system files, but it is not persistent when the virtual machine is stopped or reallocated
 - Lower read/write latency, quicker node scalability, provisioning, and cluster upgrades
 - Frequently used for stateless workloads that don't need permanent storage, like containers
 - An ephemeral OS disk, similar to the temporary drive, is included in the price of the virtual machine, saving you money on additional storage
 - The OS disk needs to fit in the VM cache if you decide to utilize an ephemeral OS
- AKS defaults to ephemeral OS, if possible, for a given node pool configuration
- Beside a few exceptions when Kubernetes pods may need persisting data to the local storage of the OS disk, like using EmptyDir or HostPath, ephemeral OS disk is recommended